

Membuat Create Read Update Delete dengan Menggunakan Excel

1. Buat folder project bernama **excel**
2. Buat file excel di dalam folder tersebut dengan nama **koleksi.xlsx**
3. Buat kolom seperti berikut ini

	A	B	C	D	E	F	G
1	ID	Kode_Buku	Judul	Pengarang	Penerbit	Tahun	Status
2							
3							
4							

4. Copy paste koding berikut dan beri nama file **index.php**

```
<?php

/* =====
PENGATURAN
===== */
$file = __DIR__ . '/koleksi.xlsx';

/* =====
CEK FILE
===== */
if (!file_exists($file)) {
    die('File koleksi.xlsx tidak ditemukan!');
}

/* =====
FUNGSI ESCAPE XML
===== */
function xml_escape($text)
{
    return htmlspecialchars(
        (string)$text,
        ENT_QUOTES,
        'UTF-8'
    );
}

/* =====
BACA DATA EXCEL
===== */

function bacaExcel($file)
{
    $zip = new ZipArchive();

    if ($zip->open($file) !== TRUE) {
        die('Gagal membuka file koleksi.xlsx');
    }

    /* -----
    Shared Strings
    ----- */
    $sharedStrings = array();
    $index = $zip->locateName('xl/sharedStrings.xml');
    if ($index !== false) {
        $xmlShared = simplexml_load_string(
            $zip->getFromIndex($index)
        );
        foreach ($xmlShared->si as $si) {
            $text = '';
            if (isset($si->t)) {
                $text = (string)$si->t;
            } else {
                foreach ($si->r as $r) {
                    $text .= (string)$r->t;
                }
            }
            $sharedStrings[] = $text;
        }
    }

    /* -----
    Sheet pertama
    ----- */

    $sheet = $zip->getFromName(
        'xl/worksheets/sheet1.xml'
    );

    if ($sheet === false) {
        $zip->close();
        die('Sheet Excel tidak ditemukan!');
    }
}
```

```

$xml = simplexml_load_string($sheet);
$data = array();

foreach ($xml->sheetData->row as $row) {
    $baris = array();
    foreach ($row->c as $cell) {
        $ref = (string)$cell['r'];
        $value = '';
        $type = (string)$cell['t'];

        /* Ambil nama kolom
        contoh A1 = A
        B1 = B
        */

        preg_match(
            '/^[A-Z]+/',
            $ref,
            $matches
        );

        if (!isset($matches[0])) {
            continue;
        }

        $kolom = $matches[0];
        /* Ubah huruf kolom menjadi nomor */
        $posisi = 0;

        for ($i = 0; $i < strlen($kolom); $i++) {

            $posisi =
                $posisi * 26 +
                (ord($kolom[$i]) - 64);
        }

        $posisi--;
        /* -----
        Ambil isi cell
        ----- */
        if ($type == 'inlineStr') {

            if (isset($cell->is->t)) {
                $value = (string)$cell->is->t;
            }

        } else {

            $value = (string)$cell->v;

        }

        /* Shared string */
        if ($type == 's') {

            $nomor = (int)$value;

            if (isset($sharedStrings[$nomor])) {
                $value = $sharedStrings[$nomor];
            } else {
                $value = '';
            }

        }

        $baris[$posisi] = $value;
    }

    /* Pastikan 7 kolom */
    for ($i = 0; $i < 7; $i++) {

        if (!isset($baris[$i])) {
            $baris[$i] = '';
        }

    }
    ksort($baris);
    $data[] = $baris;
}
$zip->close();
return $data;
}

/* =====
SIMPAN DATA KE EXCEL
===== */
function simpanExcel($file, $data)

```

```

{
    $zip = new ZipArchive();
    if ($zip->open($file) !== TRUE) {
        die('Gagal membuka file Excel untuk disimpan.');
```

}

/* -----
 Buat XML sheet baru
 ----- */

\$jumlahBaris = count(\$data);
 \$xml = '<?xml version="1.0" encoding="UTF-8" standalone="yes"?>';
 \$xml .= '<worksheet xmlns="http://schemas.openxmlformats.org/spreadsheetml/2006/main">';
 \$xml .= '<dimension ref="A1:G' . \$jumlahBaris . '"/>';
 \$xml .= '<sheetViews>';
 \$xml .= '<sheetView workbookViewId="0">';
 \$xml .= '</sheetView>';
 \$xml .= '</sheetViews>';
 \$xml .= '<sheetData>';

/* -----
 Buat setiap baris
 ----- */

```

    for ($i = 0; $i < count($data); $i++) {
        $nomorBaris = $i + 1;
        $xml .= '<row r="' . $nomorBaris . '">';
        for ($j = 0; $j < 7; $j++) {
            if (isset($data[$i][$j])) {
                $value = $data[$i][$j];
            } else {
                $value = '';
            }

            /* Nomor kolom */
            $hurufKolom = chr(65 + $j);
            $cellRef =
                $hurufKolom .
                $nomorBaris;

            /* Semua data disimpan sebagai inline string */
            $xml .= '<c r="' .
                $cellRef .
                '" t="inlineStr">';
            $xml .= '<is>';
            $xml .= '<t xml:space="preserve">' .
                xml_escape($value) .
                '</t>';
            $xml .= '</is>';
            $xml .= '</c>';
        }

        $xml .= '</row>';
    }

    $xml .= '</sheetData>';
    $xml .= '</worksheet>';

    /* -----
    Ganti sheet1.xml
    ----- */
    $zip->deleteName(
        'xl/worksheets/sheet1.xml'
    );

    $zip->addFromString(
        'xl/worksheets/sheet1.xml',
        $xml
    );

    if (!$zip->close()) {
        die('Gagal menyimpan file Excel.');
```

}

/* =====
 BACA DATA
 ===== */
 \$data = bacaExcel(\$file);

/* =====
 PROSES TAMBAH
 ===== */

```

    if (
        isset($_POST['aksi']) &&
        $_POST['aksi'] == 'tambah'
    ) {

        $id = isset($_POST['id'])
            ? trim($_POST['id'])
    
```

```

        : '';

$kode = isset($_POST['kode'])
    ? trim($_POST['kode'])
    : '';

$judul = isset($_POST['judul'])
    ? trim($_POST['judul'])
    : '';

$pengarang = isset($_POST['pengarang'])
    ? trim($_POST['pengarang'])
    : '';

$penerbit = isset($_POST['penerbit'])
    ? trim($_POST['penerbit'])
    : '';

$tahun = isset($_POST['tahun'])
    ? trim($_POST['tahun'])
    : '';

$status = isset($_POST['status'])
    ? trim($_POST['status'])
    : '';

$data[] = array(
    $id,
    $kode,
    $judul,
    $pengarang,
    $penerbit,
    $tahun,
    $status
);

simpanExcel($file, $data);
header('Location: index.php?pesan=tambah');
exit;
}

/* =====
PROSES EDIT
===== */

if (
    isset($_POST['aksi']) &&
    $_POST['aksi'] == 'edit'
) {

    $index = isset($_POST['index'])
        ? (int)$_POST['index']
        : -1;

    if ($index >= 1 && $index < count($data)) {

        $data[$index][0] =
            isset($_POST['id'])
            ? trim($_POST['id'])
            : '';

        $data[$index][1] =
            isset($_POST['kode'])
            ? trim($_POST['kode'])
            : '';

        $data[$index][2] =
            isset($_POST['judul'])
            ? trim($_POST['judul'])
            : '';

        $data[$index][3] =
            isset($_POST['pengarang'])
            ? trim($_POST['pengarang'])
            : '';

        $data[$index][4] =
            isset($_POST['penerbit'])
            ? trim($_POST['penerbit'])
            : '';

        $data[$index][5] =
            isset($_POST['tahun'])
            ? trim($_POST['tahun'])
            : '';

        $data[$index][6] =
            isset($_POST['status'])

```

```

        ? trim($_POST['status'])
        : '';

        simpanExcel($file, $data);

        header('Location: index.php?pesan=edit');
        exit;
    }
}

/* =====
PROSES HAPUS
===== */

if (
    isset($_GET['hapus'])
) {
    $index = (int)$ _GET['hapus'];
    /* Jangan hapus header */
    if ($index >= 1 && $index < count($data)) {
        array_splice(
            $data,
            $index,
            1
        );
        simpanExcel($file, $data);
        header('Location: index.php?pesan=hapus');
        exit;
    }
}

/* =====
MODE EDIT
===== */

$editData = null;
$editIndex = -1;

if (isset($_GET['edit'])) {

    $editIndex = (int)$ _GET['edit'];

    if (
        $editIndex >= 1 &&
        $editIndex < count($data)
    ) {

        $editData =
            $data[$editIndex];
    }
}

/* =====
PENCARIAN
===== */
$cari = '';
if (isset($_GET['cari'])) {
    $cari = trim($_GET['cari']);
}

/* =====
HTML
===== */

?>

<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>CRUD Koleksi Buku</title>
<style>

* {
    box-sizing: border-box;
}

body {
    font-family: Arial, sans-serif;
    background: #f3f4f6;
    margin: 0;
    padding: 30px;
    color: #333;
}

.container {
    max-width: 1200px;

```

```

        margin: auto;
    }

    h1 {
        margin-bottom: 20px;
    }

    .card {
        background: white;
        padding: 20px;
        border-radius: 8px;
        margin-bottom: 20px;
        box-shadow: 0 2px 8px rgba(0,0,0,0.08);
    }

    .form-grid {
        display: grid;
        grid-template-columns: repeat(2, 1fr);
        gap: 12px;
    }

    .form-group {
        display: flex;
        flex-direction: column;
    }

    .form-group label {
        margin-bottom: 5px;
        font-weight: bold;
    }

    .form-group input,
    .form-group select {
        padding: 10px;
        border: 1px solid #ccc;
        border-radius: 5px;
        font-size: 14px;
    }

    .full {
        grid-column: 1 / 3;
    }

    button,
    .btn {
        display: inline-block;
        padding: 9px 14px;
        border: none;
        border-radius: 5px;
        cursor: pointer;
        text-decoration: none;
        font-size: 14px;
    }

    .btn-primary {
        background: #3498db;
        color: white;
    }

    .btn-success {
        background: #27ae60;
        color: white;
    }

    .btn-warning {
        background: #f39c12;
        color: white;
    }

    .btn-danger {
        background: #e74c3c;
        color: white;
    }

    .btn-secondary {
        background: #7f7f7f;
        color: white;
    }

    .search {
        display: flex;
        gap: 10px;
    }

    .search input {
        flex: 1;
        padding: 10px;
        border: 1px solid #ccc;
    }

```

```

        border-radius: 5px;
    }

    .table-wrapper {
        overflow-x: auto;
    }

    table {
        width: 100%;
        border-collapse: collapse;
        background: white;
    }

    th {
        background: #3498db;
        color: white;
        padding: 12px 8px;
        border: 1px solid #ddd;
        white-space: nowrap;
    }

    td {
        padding: 10px 8px;
        border: 1px solid #ddd;
    }

    tr:nth-child(even) {
        background: #f8f8f8;
    }

    tr:hover {
        background: #eaf5ff;
    }

    .aksi {
        white-space: nowrap;
    }

    .alert {
        padding: 12px;
        background: #d4edda;
        color: #155724;
        border-radius: 5px;
        margin-bottom: 20px;
    }

    @media(max-width:700px) {

        body {
            padding: 15px;
        }

        .form-grid {
            grid-template-columns: 1fr;
        }

        .full {
            grid-column: 1;
        }

    }

</style>
</head>
<body>
<div class="container">
<h1>📖 Koleksi Buku</h1>

<?php
if (isset($_GET['pesan'])) {
    $pesan = '';
    if ($_GET['pesan'] == 'tambah') {
        $pesan = 'Data berhasil ditambahkan.';
    }

    if ($_GET['pesan'] == 'edit') {
        $pesan = 'Data berhasil diubah.';
    }

    if ($_GET['pesan'] == 'hapus') {
        $pesan = 'Data berhasil dihapus.';
    }

    if ($pesan != '') {
        echo '<div class="alert">';
        echo $pesan;
    }
}

```

```

        echo '</div>';
    }
}

?>

<!-- =====
FORM TAMBAH / EDIT
===== -->

<div class="card">
<?php if ($editData != null) { ?>
<h2>Edit Data Buku</h2>
<form method="post">
<input type="hidden" name="aksi" value="edit">
<input
    type="hidden"
    name="index"
    value="<?php echo $editIndex; ?>"
>

<div class="form-grid">

<div class="form-group">
<label>ID</label>
<input
    type="text"
    name="id"
    value="<?php echo htmlspecialchars(
        $editData[0],
        ENT_QUOTES,
        'UTF-8'
    ); ?>"
    required
>

</div>
<div class="form-group">
<label>Kode Buku</label>
<input
    type="text"
    name="kode"
    value="<?php echo htmlspecialchars(
        $editData[1],
        ENT_QUOTES,
        'UTF-8'
    ); ?>"
    required
>

</div>
<div class="form-group">
<label>Judul</label>
<input
    type="text"
    name="judul"
    value="<?php echo htmlspecialchars(
        $editData[2],
        ENT_QUOTES,
        'UTF-8'
    ); ?>"
    required
>

</div>
<div class="form-group">
<label>Pengarang</label>
<input
    type="text"
    name="pengarang"
    value="<?php echo htmlspecialchars(
        $editData[3],
        ENT_QUOTES,
        'UTF-8'
    ); ?>"
>

</div>
<div class="form-group">
<label>Penerbit</label>
<input
    type="text"
    name="penerbit"
    value="<?php echo htmlspecialchars(
        $editData[4],
        ENT_QUOTES,
        'UTF-8'
    ); ?>"

```

```

>

</div>

<div class="form-group">
<label>Tahun</label>
<input
    type="text"
    name="tahun"
    value="<?php echo htmlspecialchars(
        $editData[5],
        ENT_QUOTES,
        'UTF-8'
    ); ?>"
>

</div>

<div class="form-group">
<label>Status</label>
<select name="status">
<option value="Tersedia"
<?php
if ($editData[6] == 'Tersedia') {
    echo 'selected';
}
?>
>
Tersedia
</option>

<option value="Dipinjam"
<?php
if ($editData[6] == 'Dipinjam') {
    echo 'selected';
}
?>
>
Dipinjam
</option>
</select>
</div>

<div class="full">
<button
    type="submit"
    class="btn btn-success"
>
 Simpan Perubahan
</button>

<a
    href="index.php"
    class="btn btn-secondary"
>
Batal
</a>
</div>
</div>
</form>

<?php } else { ?>

<h2>Tambah Buku</h2>
<form method="post">
<input type="hidden" name="aksi" value="tambah">
<div class="form-grid">
<div class="form-group">
<label>ID</label>
<input
    type="text"
    name="id"
    required
>
</div>

<div class="form-group">
<label>Kode Buku</label>
<input
    type="text"
    name="kode"
    required
>
</div>
<div class="form-group">

```

```

<label>Judul</label>

<input
  type="text"
  name="judul"
  required
>

</div>

<div class="form-group">
<label>Pengarang</label>
<input
  type="text"
  name="pengarang"
>
</div>
<div class="form-group">

<label>Penerbit</label>

<input
  type="text"
  name="penerbit"
>

</div>

<div class="form-group">
<label>Tahun</label>
<input
  type="text"
  name="tahun"
>
</div>

<div class="form-group">
<label>Status</label>
<select name="status">
<option value="Tersedia">
Tersedia
</option>

<option value="Dipinjam">
Dipinjam
</option>
</select>
</div>
<div class="full">
<button
  type="submit"
  class="btn btn-primary"
>
✚ Tambah Buku
</button>
</div>
</div>
</form>
<?php } ?>
</div>

<!-- =====
PENCARIAN
===== -->

<div class="card">

<form method="get" class="search">

<input
  type="text"
  name="cari"
  placeholder="Cari ID, kode, judul, pengarang, penerbit..."
  value="<?php echo htmlspecialchars(
    $cari,
    ENT_QUOTES,
    'UTF-8'
  ); ?>"
>

<button
  type="submit"
  class="btn btn-primary"
>
🔍 Cari

```

```

</button>

<a
    href="index.php"
    class="btn btn-secondary"
>
Reset
</a>
</form>
</div>

<!-- =====
TABEL DATA
===== -->

<div class="card">

<div class="table-wrapper">

<table>

<thead>

<tr>

<th>ID</th>
<th>Kode Buku</th>
<th>Judul</th>
<th>Pengarang</th>
<th>Penerbit</th>
<th>Tahun</th>
<th>Status</th>
<th>Aksi</th>
</tr>
</thead>
<tbody>

<?php
$ada = false;
/* Mulai dari 1 karena baris 0 adalah header */
for ($i = 1; $i < count($data); $i++) {
    $baris = $data[$i];
    /* -----
    Filter pencarian
    ----- */
    if ($cari != '') {
        $gabungan =
            $baris[0] . ' ' .
            $baris[1] . ' ' .
            $baris[2] . ' ' .
            $baris[3] . ' ' .
            $baris[4] . ' ' .
            $baris[5] . ' ' .
            $baris[6];

        if (
            strpos(
                $gabungan,
                $cari
            ) === false
        ) {
            continue;
        }
    }

    $ada = true;
    echo '<tr>';

    for ($j = 0; $j < 7; $j++) {
        echo '<td>';
        echo htmlspecialchars(
            $baris[$j],
            ENT_QUOTES,
            'UTF-8'
        );
        echo '</td>';
    }
    echo '<td class="aksi">';

    /* Tombol Edit */
    echo '<a
        href="index.php?edit=' . $i . '"
        class="btn btn-warning"
    >';
    echo '✎ Edit';
    echo '</a> ';

```

```

/* Tombol Hapus */

echo '<a
    href="index.php?hapus=' . $i . '"
    class="btn btn-danger"
    onclick="return confirm(\'Yakin ingin menghapus data ini?\');"
>';
echo '✖ Hapus';
echo '</a>';
echo '</td>';
echo '</tr>';
}

if (!$ada) {
    echo '<tr>';
    echo '<td colspan="8" style="text-align:center;">';
    echo 'Data tidak ditemukan.';
    echo '</td>';
    echo '</tr>';
}

?>

</tbody>
</table>
</div>
</div>
</div>
</body>
</html>

```

5. Selamat, project berhasil dibuat

localhost:8123/excel/

 **Koleksi Buku**

Tambah Buku

ID

Kode Buku

Judul

Pengarang

Penerbit

Tahun

Status

Tersedia ▼

+ Tambah Buku

Cari

Reset

ID	Kode Buku	Judul	Pengarang	Penerbit	Tahun	Status	Aksi
Data tidak ditemukan.							